DISTINCT MONOID CHARACTERS INTO A FIELD ARE LINEARLY INDEPENDENT (ARTIN'S THEOREM)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The coefficient-vanishing branch is compressed by the author into 'Since lambda_i not = 0 we conclude that chi_n(h) = chi_i(h) for all i by induction' (source lines 1709-1711): the displayed relation holds for all g, so induction on the n-1 distinct characters forces each coefficient lambda_i (chi_n(h) - chi_i(h)) to vanish.

Editorial note. The label lemma-independence-characters names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Monoid homomorphisms G -> L (L regarded as a monoid under multiplication) send the neutral element of G to 1
- Same chapter use: the Artin-Schreier and power-sum arguments both invoke this lemma (not part of the counted unit)

Lemma 0.1. *Let L be a field. Let G be a monoid, for example a group. Let $\chi_1, \ldots, \chi_n : G \to L$ be pairwise distinct homomorphisms of monoids where L is regarded as a monoid by multiplication. Then $\chi_1, \ldots, \chi_n$ are L-linearly independent: if $\lambda_1, \ldots, \lambda_n \in L$ not all zero, then $\sum \lambda_i \chi_i(g) \neq 0$ for some $g \in G$.*

Proof. If $n = 1$ this is true because $\chi_1(e) = 1$ if $e \in G$ is the neutral (identity) element. We prove the result by induction for $n > 1$. Suppose that $\lambda_1, \ldots, \lambda_n \in L$ not all zero. If $\lambda_i = 0$ for some, then we win by induction on n. Since we want to show that $\sum \lambda_i \chi_i(g) \neq 0$ for some $g \in G$ we may after dividing by $-\lambda_n$ assume that $\lambda_n = -1$. Then the only way we get in trouble is if

$$\chi_n(g) = \sum\nolimits_{i=1,\ldots,n-1} \lambda_i \chi_i(g)$$

for all $g \in G$. Fix $h \in G$. Then we would also get

$$\begin{aligned} \chi_n(h)\chi_n(g) &= \chi_n(hg) \\ &= \sum\nolimits_{i=1,\ldots,n-1} \lambda_i \chi_i(hg) \\ &= \sum\nolimits_{i=1,\ldots,n-1} \lambda_i \chi_i(h)\chi_i(g) \end{aligned}$$

Multiplying the previous relation by $\chi_n(h)$ and subtracting we obtain

$$0 = \sum\nolimits_{i=1,\ldots,n-1} \lambda_i (\chi_n(h) - \chi_i(h))\chi_i(g)$$

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for all $g \in G$. Since $\lambda_i \neq 0$ we conclude that $\chi_n(h) = \chi_i(h)$ for all i by induction. The choice of h above was arbitrary, so we conclude that $\chi_i = \chi_n$ for $i \leq n-1$ which contradicts the assumption that our characters χ_i are pairwise distinct. $\square$